

DNA/Protein Sequence Analysis Using BLASTP

CodeAlpha Bioinformatics Internship – Task 1

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1. Introduction

Bioinformatics combines biology, computer science, and statistics to analyze biological data. One of the most widely used bioinformatics tools is the Basic Local Alignment Search Tool (BLAST), developed by the National Center for Biotechnology Information (NCBI). BLAST compares a query sequence against biological databases to identify homologous sequences and estimate evolutionary relationships.

In this project, the amino acid sequence of the human hemoglobin beta (HBB) protein was obtained from UniProt and analyzed using BLASTP. The objective was to identify homologous proteins, compare sequence similarity, and interpret the biological significance of the obtained results.

2. Materials and Methods

Query Sequence

The protein selected for this analysis was **Human Hemoglobin Subunit Beta (HBB)**.

Database: UniProt

UniProt ID: P68871

Protein length: 147 amino acids

Software and Database

- UniProt Knowledgebase
- NCBI BLASTP
- Database: ClusteredNR (Clustered Non-Redundant Protein Database)

Procedure

1. The amino acid sequence of human hemoglobin beta was downloaded from UniProt in FASTA format.
2. The sequence was submitted to the NCBI BLASTP server.
3. The ClusteredNR database was selected for searching homologous proteins.
4. Significant hits were evaluated based on:
 - Bit Score
 - E-value
 - Percentage Identity
 - Query Coverage
 - Sequence Alignment

3. Results

The BLASTP search identified numerous homologous hemoglobin proteins from different mammalian species. The obtained results demonstrate that hemoglobin beta is highly conserved among vertebrates.

Table 1. Top BLASTP Matches

Protein	Organism	Identity	Similarity (Positives)	Score	E-value
Hb Monza protein	Homo sapiens	86%	86%	288 bits	2e-97
Beta-globin	Homo sapiens	93%	93%	277 bits	2e-93
Hemoglobin beta-like	Canis lupus familiaris	90%	93%	275 bits	1e-92
Hemoglobin delta	Ailuropoda melanoleuca	90%	93%	273 bits	6e-92
Hemoglobin beta	Rhinonycteris naso	89%	92%	271 bits	2e-91

Alignment Analysis

The best alignment was obtained with **Hb Monza protein (Homo sapiens)**.

Alignment statistics:

- Score: **288 bits**
- E-value: **2×10^{-97}**
- Identities: **147/170 (86%)**
- Positives: **147/170 (86%)**
- Gaps: **23/170 (13%)**
-

The second-best alignment corresponded to another human beta-globin protein sequence.

Statistics:

- Score: **277 bits**
- E-value: **2×10^{-93}**
- Identities: **137/147 (93%)**
- Positives: **137/147 (93%)**
- Gaps: **0**

Job Title

sp|P68871|HBB_HUMAN Hemoglobin subunit beta...

RID

4TACR594014

Search expires on 07-08 15:24 pm

Download All

Program

BLASTP

Citation

Database

ClusteredNR

See details

Query ID

Icl|Query_3257333

Description

sp|P68871|HBB_HUMAN Hemoglobin subunit beta OS=Ho ...

Molecule type

amino acid

Query Length

147

Other reports

Distance tree of results

Multiple alignment

MSA viewer

Filter Results

Organism

only top 20 will appear

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity

to

E value

to

Query Coverage

to

Filter

Reset

Clusters

Graphic Summary

Alignments

Taxonomy

Clusters producing significant alignments

Download

Select columns

Show

100

select all

100 clusters selected

GenPept

Graphics

Distance tree of results

Multiple alignment

MSA Viewer

Cluster Composition

Cluster Ancestor

Cluster Representative Sequence

Max Score

Total Score

Query Cover

E value

Per. Ident

Acc. Len

Accession

Click the to see the cluster contents

1 member(s), 1 organism(s)

human

Hb Monza protein [Homo sapiens]

288

288

100%

2e-97

86.47%

170

XEF57200.1

1 member(s), 1 organism(s)

human

beta-globin [Homo sapiens]

277

277

100%

2e-93

93.20%

147

AAA35597.1

9 member(s), 7 organism(s)

dog_coyote_wolf_fox

hemoglobin subunit beta-like [Canis lupus familiaris]

275

275

100%

1e-92

89.80%

147

NP_001257812.1

184 member(s), 76 organism(s)

placentals

hemoglobin subunit delta [Ailuropoda melanoleuca]

273

273

100%

6e-92

89.80%

147

NP_001291814.1

18 member(s), 17 organism(s)

placentals

hemoglobin subunit beta-like [Ailuropoda melanoleuca]

273

273

100%

7e-92

89.80%

147

NP_001291812.1

9 member(s), 6 organism(s)

bats

hemoglobin subunit beta [Rhynchonycteris naso]

271

271

100%

2e-91

89.12%

147

KAM8815943.1

1 member(s), 1 organism(s)

giant panda

hypothetical protein PANDA_015769 [Ailuropoda melanoleuca]

275

446

100%

3e-91

90.48%

258

EFB18430.1

Figure 1. BLASTP results table showing the highest-scoring homologous protein sequences identified for the human hemoglobin beta (HBB) protein. The best matches include human hemoglobin variants and homologous proteins from other mammalian species.

Alignment view	Pairwise	Restore defaults	Download ▼
100 clusters selected ?			
Download ▼ GenPept Graphics Next Previous Descriptions			
Hb Monza protein [Homo sapiens] Sequence ID: XEF57200.1 Length: 170 Number of Matches: 1			
Range 1: 1 to 170 GenPept Graphics Next Match Previous Match			
Score	Expect	Method	Identities
288 bits(737)	2e-97	Compositional matrix adjust.	147/170(86%)
Positives	147/170(86%)		
Gaps	23/170(13%)		
Query 1	MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPK	60	
Sbjct 1	MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPK	60	
Query 61	VKAHGKKVLGAFSDGLAHLDN-----LKGTFATLSSELHCDKL	97	
Sbjct 61	VKAHGKKVLGAFSDGLAHLDNPKVKAHGKKVLGAFSDGLAHLDNLKGTFATLSSELHCDKL	120	
Query 98	HVDPENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH	147	
Sbjct 121	HVDPENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH	170	

Figure 2. Pairwise BLASTP sequence alignment between the query human hemoglobin beta (HBB) protein and the best matching homologous protein (Hb Monza, *Homo sapiens*). The alignment shows a Bit Score of 288, an E-value of 2×10^{-97} , 86% sequence identity, and 13% gaps.

4. Discussion

The BLASTP analysis revealed a high degree of conservation of the hemoglobin beta protein among mammals. High sequence identities (82–93%) and extremely low E-values indicate that these proteins share a common evolutionary origin.

The best matches were proteins from humans, followed by dogs, pandas, bats, pigs, hippopotamuses, rodents, and other mammals. Such conservation is expected because hemoglobin performs the essential biological function of oxygen transport in vertebrates.

The Bit Scores were consistently high (>250 bits), confirming highly significant alignments. Likewise, E-values approaching zero indicate that the observed similarities are not due to random chance.

Minor sequence differences between species likely reflect evolutionary adaptation while preserving the protein's biological function.

5. Conclusion

This study successfully identified homologous hemoglobin beta proteins using BLASTP. The analysis demonstrated that human hemoglobin beta is highly conserved among mammalian species.

Key findings include:

- High sequence similarity among mammals.
- Extremely significant BLAST matches (very low E-values).
- Strong evolutionary conservation of the hemoglobin beta protein.
- Reliable identification of homologous sequences using BLASTP.

Overall, BLAST proved to be an effective bioinformatics tool for protein sequence comparison and evolutionary analysis.

References

1. UniProt Consortium. *UniProt Knowledgebase (UniProtKB)*.
<https://www.uniprot.org/>
2. National Center for Biotechnology Information (NCBI). *BLAST: Basic Local Alignment Search Tool*. <https://blast.ncbi.nlm.nih.gov/>
3. Altschul SF et al. (1990). **Basic Local Alignment Search Tool**. *Journal of Molecular Biology*, 215(3), 403–410.